

A Short Guide to Proposition C.4

The orbital Gaussian flat-prior upper bound

Abstract

Proposition C.4 proves that no channel can amplify the displaced thermal Gaussian model with asymptotic flat-prior fidelity larger than $B_{\text{orb}}(\gamma, p)$. Its proof averages a channel over a large phase-space cube, extracts an exactly covariant trace-nonincreasing limit using only compact observables, and evaluates a specially chosen weighted moment. The compact witness avoids any need to pass fidelity itself through the weak limit. This note explains the argument and how Lemma C.1, Lemma C.2, and Proposition C.3 fit together.

1 The Gaussian problem

The orbital LAN model consists of displaced product thermal states on a finite-mode Fock space $\mathcal{F}_{\text{orb}}$:

$$\Phi_z = D(z)\tau D(z)^*, \quad \Psi_z = D(\sqrt{\gamma}z)\tau D(\sqrt{\gamma}z)^*.$$

The notation suppresses that the target has the same centered thermal state τ as the ideal product model, whereas a physical amplifier necessarily produces the noisier centered state $\tau^{(\gamma)}$. The product quantum-limited amplifier sends

$$\Phi_z \longmapsto D(\sqrt{\gamma}z)\tau^{(\gamma)} D(\sqrt{\gamma}z)^*.$$

Its fidelity with Ψ_z is independent of z and equals

$$B(\tau^{(\gamma)}, \tau) = B_{\text{orb}}(\gamma, p).$$

For the cube

$$Z_L = \{z : |\Re z_\ell|, |\Im z_\ell| \leq L\},$$

Proposition C.4 supplies the converse inequality

$$\limsup_{L \rightarrow \infty} \sup_{\Lambda} \frac{1}{|Z_L|} \int_{Z_L} B(\Lambda(\Phi_z), \Psi_z) dz \leq B_{\text{orb}}(\gamma, p). \quad (1)$$

2 The witness and its three properties

Write the two centered thermal states in their joint number basis:

$$\tau = \sum_{\mathbf{k}} t_{\mathbf{k}} |\mathbf{k}\rangle\langle\mathbf{k}|, \quad \tau^{(\gamma)} = \sum_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)} |\mathbf{k}\rangle\langle\mathbf{k}|.$$

Define

$$W_\gamma = \sum_{\mathbf{k}} \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} |\mathbf{k}\rangle\langle\mathbf{k}|. \quad (2)$$

This witness is chosen so that three otherwise separate parts of the proof match exactly.

1. Its eigenvalue at $\mathbf{k} = (k_{ij})_{i < j}$ is

$$\sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} = C_{\gamma} \prod_{i < j} r_{ij}^{k_{ij}}, \quad C_{\gamma} = \prod_{i < j} \sqrt{\frac{1 - q_{ij}}{1 - q_{ij}^{(\gamma)}}}, \quad r_{ij} = \sqrt{\frac{q_{ij}}{q_{ij}^{(\gamma)}}}.$$

Because $\gamma > 1$, $q_{ij}^{(\gamma)} > q_{ij}$ and therefore $0 < r_{ij} < 1$. Hence these eigenvalues lie in $(0, C_{\gamma}]$ and converge to zero as $|\mathbf{k}| = \sum_{i < j} k_{ij} \rightarrow \infty$; this proves directly that W_{γ} is bounded, positive, injective, and compact.

2. It is an admissible product of decreasing photon-number weights for the least-noise moment in Proposition C.3.

3. It satisfies

$$\text{Tr}(\tau^{(\gamma)} W_{\gamma}) = \text{Tr}(\tau W_{\gamma}^{-1}) = \sum_{\mathbf{k}} \sqrt{t_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}} = B_{\text{orb}}(\gamma, p). \quad (3)$$

Lemma C.1, the weighted fidelity inequality, therefore gives for every positive trace-class R

$$B(R, \tau)^2 \leq \text{Tr}(R W_{\gamma}) B_{\text{orb}}(\gamma, p). \quad (4)$$

The nonlinear fidelity has been reduced to one linear compact-observable moment.

3 Statement of Proposition C.4

Proposition C.4 (rephrased). *For the orbital Gaussian input and target families,*

$$\limsup_{L \rightarrow \infty} \sup_{\Lambda} \int_{Z_L} B(\Lambda(\Phi_z), \Psi_z) \frac{dz}{|Z_L|} \leq B_{\text{orb}}(\gamma, p),$$

where the supremum is over all channels on the orbital Fock space.

4 Proof walkthrough

Step 1: move every displacement back to the origin

For each z , center the input and output of Λ by defining the channel

$$\Lambda_z(X) = D(-\sqrt{\gamma}z) \Lambda(D(z) X D(z)^*) D(-\sqrt{\gamma}z)^*.$$

The input conjugation inserts the displacement z , while the output conjugation removes the target displacement $\sqrt{\gamma}z$. In particular, using $\Phi_z = D(z) \tau D(z)^*$ and $\Psi_z = D(\sqrt{\gamma}z) \tau D(\sqrt{\gamma}z)^*$, unitary invariance of root fidelity gives

$$B(\Lambda(\Phi_z), \Psi_z) = B(\Lambda_z(\tau), \tau).$$

Now average these centered channels over the cube:

$$\Gamma_{\Lambda, L} := \frac{1}{|Z_L|} \int_{Z_L} \Lambda_z \, dz. \quad (5)$$

For each trace-class input X , this means a Bochner integral of $z \mapsto \Lambda_z(X)$ in trace norm. Strong continuity of Weyl conjugation and boundedness of the channels make that integral well defined;

later channel-valued integrals are understood pointwise in the same way. Each Λ_z is a channel, so their average $\Gamma_{\Lambda,L}$ is also a channel. Concavity of root fidelity in its first argument now gives

$$\begin{aligned} \frac{1}{|Z_L|} \int_{Z_L} B(\Lambda(\Phi_z), \Psi_z) dz &= \frac{1}{|Z_L|} \int_{Z_L} B(\Lambda_z(\tau), \tau) dz \\ &\leq B\left(\frac{1}{|Z_L|} \int_{Z_L} \Lambda_z(\tau) dz, \tau\right) \\ &= B(\Gamma_{\Lambda,L}(\tau), \tau). \end{aligned} \tag{6}$$

Thus all parameter values are collected into the performance of one averaged channel at the centered input.

Step 2: the averaged maps become covariant

Define the payoff of a channel on the cube of radius L by

$$F_L(\Lambda) = \frac{1}{|Z_L|} \int_{Z_L} B(\Lambda(\Phi_z), \Psi_z) dz, \quad F_* := \limsup_{L \rightarrow \infty} \sup_{\Lambda} F_L(\Lambda).$$

By the definition of the limsup, choose $L_j \rightarrow \infty$ such that $\sup_{\Lambda} F_{L_j}(\Lambda) \rightarrow F_*$. The supremum need not be attained, so for each j choose a channel Λ_j satisfying $F_{L_j}(\Lambda_j) \geq \sup_{\Lambda} F_{L_j}(\Lambda) - 1/j$. Then $F_{L_j}(\Lambda_j) \rightarrow F_*$. Let $\Lambda_{j,z}$ be the centered channel constructed from Λ_j in Step 1 and set $\Gamma_j := \Gamma_{\Lambda_j, L_j}$. Substituting $L = L_j$ and $\Lambda = \Lambda_j$ into (6), and using the definition of Γ_j , gives

$$F_{L_j}(\Lambda_j) = \frac{1}{|Z_{L_j}|} \int_{Z_{L_j}} B(\Lambda_j(\Phi_z), \Psi_z) dz \leq B(\Gamma_j(\tau), \tau).$$

Since the left side converges to F_* , it follows that $F_* \leq \liminf_j B(\Gamma_j(\tau), \tau)$. Thus it is enough to obtain the desired upper bound for the centered fidelities on the right.

To derive the reindexing identity, recall that $\text{Ad}_{D(a)} \circ \text{Ad}_{D(b)} = \text{Ad}_{D(a+b)}$: although $D(a)D(b)$ may contain a scalar Weyl phase, that phase cancels in UYU^* . Starting from the definition of the centered channel, we therefore obtain

$$\begin{aligned} \Lambda_{j,z}(D(w)XD(w)^*) &= \text{Ad}_{D(-\sqrt{\gamma}z)}\left(\Lambda_j\left(\text{Ad}_{D(z+w)}(X)\right)\right) \\ &= \text{Ad}_{D(\sqrt{\gamma}w)}\left(\text{Ad}_{D(-\sqrt{\gamma}(z+w))}\left(\Lambda_j\left(\text{Ad}_{D(z+w)}(X)\right)\right)\right) \\ &= D(\sqrt{\gamma}w)\Lambda_{j,z+w}(X)D(\sqrt{\gamma}w)^*. \end{aligned}$$

The first line combines the two input displacements into $z + w$. The second uses $-\sqrt{\gamma}z = \sqrt{\gamma}w - \sqrt{\gamma}(z + w)$, thereby separating the output displacement $\sqrt{\gamma}w$ from the centering displacement at $z + w$. Integrating this identity gives

$$\begin{aligned} \Gamma_j(D(w)XD(w)^*) &= D(\sqrt{\gamma}w) \left[\frac{1}{|Z_{L_j}|} \int_{Z_{L_j}+w} \Lambda_{j,u}(X) du \right] D(\sqrt{\gamma}w)^*, \\ D(\sqrt{\gamma}w)\Gamma_j(X)D(\sqrt{\gamma}w)^* &= D(\sqrt{\gamma}w) \left[\frac{1}{|Z_{L_j}|} \int_{Z_{L_j}} \Lambda_{j,u}(X) du \right] D(\sqrt{\gamma}w)^*. \end{aligned}$$

The integrals agree on the overlap of the two cubes. Since every $\Lambda_{j,u}$ is a channel and hence $\|\Lambda_{j,u}(X)\|_1 \leq \|X\|_1$, their difference satisfies

$$\begin{aligned} &\|\Gamma_j(D(w)XD(w)^*) - D(\sqrt{\gamma}w)\Gamma_j(X)D(\sqrt{\gamma}w)^*\|_1 \\ &\leq \frac{|(Z_{L_j} + w) \triangle Z_{L_j}|}{|Z_{L_j}|} \|X\|_1. \end{aligned} \tag{7}$$

Here $\triangle$ denotes symmetric difference:

$$(Z_{L_j} + w) \triangle Z_{L_j} = ((Z_{L_j} + w) \setminus Z_{L_j}) \cup (Z_{L_j} \setminus (Z_{L_j} + w)).$$

It is precisely the set of points belonging to one cube but not the other. Thus only the boundary portion on which the shifted and unshifted cubes differ contributes. More concretely, if $\|w\|_{1,\mathbb{R}} = \sum_\ell(|\Re w_\ell| + |\Im w_\ell|)$, then for fixed w and all sufficiently large L ,

$$\frac{|(Z_L + w) \triangle Z_L|}{|Z_L|} \leq \frac{\|w\|_{1,\mathbb{R}}}{L} \longrightarrow 0.$$

This is the Følner property of the cubes and proves that Γ_j becomes asymptotically displacement covariant.

Each Γ_j is a channel, so $\|\Gamma_j\|_{1 \rightarrow 1} = 1$. Lemma C.2 therefore yields a subsequence and a completely positive trace-nonincreasing map Γ such that

$$\mathrm{Tr}[\Gamma_j(X)K] \longrightarrow \mathrm{Tr}[\Gamma(X)K] \quad (K \text{ compact}). \quad (8)$$

To see explicitly how (7) and (8) give exact covariance, abbreviate $U_w = D(\sqrt{\gamma}w)$ and define the covariance defect

$$E_j(X, w) := \Gamma_j(D(w)XD(w)^*) - U_w\Gamma_j(X)U_w^*.$$

Equation (7) says

$$\|E_j(X, w)\|_1 \leq \frac{|(Z_{L_j} + w) \triangle Z_{L_j}|}{|Z_{L_j}|} \|X\|_1 \longrightarrow 0.$$

Hence, for every compact K , the trace Hölder inequality gives

$$|\mathrm{Tr}(E_j(X, w)K)| \leq \|E_j(X, w)\|_1 \|K\| \longrightarrow 0.$$

Now apply (8) to the two terms defining $E_j(X, w)$. For the first term,

$$\mathrm{Tr}[\Gamma_j(D(w)XD(w)^*)K] \longrightarrow \mathrm{Tr}[\Gamma(D(w)XD(w)^*)K].$$

For the second term, $U_w^*KU_w$ is compact, and therefore

$$\begin{aligned} \mathrm{Tr}[U_w\Gamma_j(X)U_w^*K] &= \mathrm{Tr}[\Gamma_j(X)U_w^*KU_w] \\ &\longrightarrow \mathrm{Tr}[\Gamma(X)U_w^*KU_w] = \mathrm{Tr}[U_w\Gamma(X)U_w^*K]. \end{aligned}$$

Subtracting these two limits yields

$$\mathrm{Tr}(E_j(X, w)K) \longrightarrow \mathrm{Tr}[(\Gamma(D(w)XD(w)^*) - U_w\Gamma(X)U_w^*)K].$$

The left side also tends to zero by (7), so

$$\mathrm{Tr}[(\Gamma(D(w)XD(w)^*) - U_w\Gamma(X)U_w^*)K] = 0$$

for every compact K . Compact operators separate trace-class operators, so the operator in parentheses is zero. Thus Γ is exactly displacement covariant with gain $\sqrt{\gamma}$.

The limit may lose trace because oscillator mass can escape to high energy, and the convergence above cannot test the identity operator, which is not compact. Thus one obtains only $\mathrm{Tr} \Gamma(X) \leq \mathrm{Tr} X$ for $X \geq 0$, but this causes no problem because Proposition C.3 allows trace-nonincreasing maps.

Step 3: pass the compact witness moment to the limit

Apply (4) to $R = \Gamma_j(\tau)$:

$$B(\Gamma_j(\tau), \tau)^2 \leq \text{Tr}[\Gamma_j(\tau)W_\gamma] B_{\text{orb}}(\gamma, p).$$

Since W_γ is compact, (8) gives

$$\text{Tr}[\Gamma_j(\tau)W_\gamma] \longrightarrow \text{Tr}[\Gamma(\tau)W_\gamma]. \quad (9)$$

This is the key reason for introducing the weighted witness before taking a limit. Indeed, applying the weighted inequality before taking the limit gives directly

$$\limsup_j B(\Gamma_j(\tau), \tau)^2 \leq \left(\lim_j \text{Tr}[\Gamma_j(\tau)W_\gamma] \right) B_{\text{orb}}(\gamma, p) = \text{Tr}[\Gamma(\tau)W_\gamma] B_{\text{orb}}(\gamma, p).$$

The convergence supplied by Lemma C.2 controls linear expressions $R \mapsto \text{Tr}(RK)$ only when K is compact; it does not directly control the nonlinear function $R \mapsto B(R, \tau)$. A direct passage from $\Gamma_j(\tau)$ to $\Gamma(\tau)$ inside the fidelity would therefore require a separate upper-semicontinuity result. The compact witness avoids that issue by replacing fidelity with the convergent linear moment before the limit is taken.

Step 4: use the least-noise moment

The limiting map has exactly the hypotheses of Proposition C.3. Therefore

$$\begin{aligned} \text{Tr}[\Gamma(\tau)W_\gamma] &\leq \text{Tr}[\Gamma(\tau)] \text{Tr}[\tau^{(\gamma)}W_\gamma] \\ &\leq B_{\text{orb}}(\gamma, p), \end{aligned}$$

where the last line uses $\text{Tr} \Gamma(\tau) \leq 1$ and (3). Combining this with (4)–(9) gives

$$\limsup_j B(\Gamma_j(\tau), \tau)^2 \leq B_{\text{orb}}(\gamma, p)^2.$$

Root fidelity is nonnegative, so taking square roots and returning to (6) proves (1).

5 Why the argument is sharp

The witness is not an arbitrary estimate. Its eigenvalue at $\mathbf{k}$ is the square-root likelihood ratio between the ideal thermal law and the quantum-limited output law. Concretely, for the quantum-limited centered output $R = \tau^{(\gamma)}$, the two factors in Lemma C.1 are

$$\text{Tr}(\tau^{(\gamma)}W_\gamma) = \sum_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)} \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} = \sum_{\mathbf{k}} \sqrt{t_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}}$$

and

$$\text{Tr}(\tau W_\gamma^{-1}) = \sum_{\mathbf{k}} t_{\mathbf{k}} \sqrt{\frac{t_{\mathbf{k}}^{(\gamma)}}{t_{\mathbf{k}}}} = \sum_{\mathbf{k}} \sqrt{t_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}}.$$

Thus both factors equal the classical Hellinger overlap $B_{\text{orb}}(\gamma, p)$. Since τ and $\tau^{(\gamma)}$ are diagonal in the same basis, their root fidelity is that same overlap, and hence

$$\begin{aligned} B(\tau^{(\gamma)}, \tau)^2 &= \left(\sum_{\mathbf{k}} \sqrt{t_{\mathbf{k}} t_{\mathbf{k}}^{(\gamma)}} \right)^2 \\ &= B_{\text{orb}}(\gamma, p)^2 \\ &= \text{Tr}(\tau^{(\gamma)} W_{\gamma}) \text{Tr}(\tau W_{\gamma}^{-1}). \end{aligned}$$

Therefore the two sides of the weighted Cauchy–Schwarz inequality are equal when $R = \tau^{(\gamma)}$. The least-noise proposition then says that no admissible covariant limit can have a larger expectation of this witness.

The product quantum-limited amplifier attains $B_{\text{orb}}(\gamma, p)$ at every z , so Proposition C.4 supplies the matching upper bound used in the orbital specialization of the Gaussian amplification theorem.

6 Logical dependencies

The proof has the compact chain

$$\begin{aligned} &\text{Følner averaging and concavity} \\ &\Downarrow \\ &\text{Lemma C.2: compact-observable covariant limit} \\ &\Downarrow \\ &\text{Lemma C.1: fidelity} \leq \text{witness moment} \\ &\Downarrow \\ &\text{Proposition C.3: least-noise moment} \\ &\Downarrow \\ &B_{\text{orb}}(\gamma, p). \end{aligned}$$

The proof is entirely orbital: there is no classical spectral coordinate. Proposition C.6 repeats this architecture with an additional classical witness and convolution argument.